



Understanding salt diffusion in dairy-based systems: A model approach using rennet-coagulated micellar casein concentrates

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ABSTRACT

Understanding salt diffusion is essential for optimizing salt uptake during cheese production. Although multiple interacting factors complicate this process, the utilization of a model cheese system offers a controlled platform to evaluate their relative influence. This study developed a model using renneted gels prepared from micellar casein concentrate and investigated the use of salty whey, a secondary byproduct of Cheddar production, as an alternative salting medium. We assessed the effects of varying the pH, fat, and calcium contents of the model cheese and salting temperature on salt migration. Salt diffusivity was modeled using Fick's second law. Increasing calcium content during formulation of the model system (0%–1% wt/wt, weight basis) significantly increased salt penetration. Similarly, salting at 30°C significantly enhanced salt diffusion compared with salting at 20°C or 40°C. Diffusion coefficients of salt in the model cheese systems ranged from 3.1×10^{-9} to 8.5×10^{-9} m²/s. Increasing calcium level promoted a more continuous compact protein matrix, which in turn increased salt diffusion; in contrast, fat globules hindered salt movement within the matrix. Analysis of Fourier transform infrared spectra indicated that higher levels of salt migration corresponded to more ordered protein secondary structures with stronger hydrogen bonding. The diffusivity of salty whey, however, was less than that of brine, likely due to the higher osmotic pressure and viscosity of the former.

Key words: Cheddar cheese, micellar casein concentrate, salty whey, salt diffusion coefficient

INTRODUCTION

Salting plays a critical role in cheese quality, as it affects cheese composition, microflora, and enzyme activity (Fox and Cogan, 2004; Guinee, 2004). The salt content

in different cheese types varies, ranging from as low as 0.7% (wt/wt) in Swiss-type cheeses to as high as 6% (wt/wt) in Domiati cheese (Guinee, 2004). Beyond its fundamental roles, salt concentration influences a wide range of biochemical changes in cheese, such as proteolysis, glycolysis, lipolysis, and para-casein hydration, which occur during ripening (Guinee, 2004; McCarthy et al., 2015; Bae et al., 2017).

Salt diffusion refers to the migration of salt molecules from a region of higher concentration (i.e., the salting medium) to a region of lower concentration (i.e., the cheese matrix), driven by random molecular motion and leading to uniform solute distribution (Fick, 1995). Most cheese varieties are salted via surface diffusion methods such as brining, immersion, or dry salting (Morris et al., 1985). Salt diffusivity has been extensively studied in various cheese types, including Cheddar, Gouda, Mozzarella, and white cheese (Morris et al., 1985; Turhan and Kaletunc, 1992; Luo et al., 2013). The rate and extent of salt diffusion are influenced by multiple factors, including salting temperature and duration, curd geometry, initial moisture content, salt/moisture ratio, and curd pH (Geurts et al., 1981; Melilli et al., 2003; Fox et al., 2004). These variables complicate the understanding and control of salt transport phenomena in real cheese matrices. Understanding the mechanisms underlying salt diffusion is essential for optimizing salt uptake in cheese manufacturing. This will enable better control over the salting duration, salt absorption kinetics, and salt distribution, as well as reducing the generation of byproducts such as salty or white whey.

Model cheese systems have been widely used to investigate diffusion phenomena. For instance, in a previous study, we developed a rennet-coagulated micellar casein concentrate (MCC) gel model system containing 7.5% to 15% (wt/wt) protein, without added fat, and used a standard brine as the salting medium (Alehosseini et al., 2023). Floury et al. (2009) also used a model system made from milk protein concentrate powder and anhydrous milk fat to simulate salt diffusion in cheese, and Lauerjat et al. (2009) studied the release of salt and aro-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

ma compounds in model cheeses made from ultrafiltered skim milk retentate powder. These simplified systems are particularly advantageous because real cheese matrices are complex and difficult to manipulate due to the presence of multiple interacting variables (Alehosseini et al., 2025). In contrast, model systems, such as those based on MCC gels formed by rennet coagulation, allow controlled variation of individual parameters, providing clearer insights into the relative effects of each factor on salt diffusion (Floury et al., 2009).

Salty whey is generated during cheese production (specifically Cheddar types) due to the expulsion of whey from the curd in response to osmotic pressure gradients created by high external NaCl concentrations (Weerasingha et al., 2025). This interaction between NaCl and the casein matrix causes water and solutes to be released, forming salty whey. In Cheddar cheese production, particularly during the mellowing and pressing stages, up to 50% to 60% of the added salt can be lost into the salty whey stream (Talebi et al., 2019). Despite its high nutrient content, the disposal of salty whey poses serious environmental challenges due to its high salinity and organic load, and discarding salty whey onto land or into aquatic ecosystems could contribute to substantial ecological degradation.

In Cheddar cheese manufacture, although salt is added in solid form (“dry salting”), the salt crystals dissolve on the moist surfaces of the milled curd particles, forming a thin layer of brine. Salt subsequently diffuses into the curd matrix through this aqueous phase; consequently, during dry salting, diffusion occurs via a brine phase at the curd surface. For this reason, the salting mechanism represented in the current MCC model more closely resembles that of Cheddar-type cheese salting rather than complete immersion brining used for cheeses such as Mozzarella. Further, the present study advances the research conducted by Alehosseini et al. (2023) by formulating MCC gels with a higher protein content (20% wt/wt) and variable fat levels to better mimic the structure of Cheddar cheese. Furthermore, salty whey was also investigated as an alternative salting medium. Therefore, this study aimed to evaluate: (1) salt diffusion mechanisms using a rennet-induced MCC model system, (2) the relative effect of varying pH, fat and calcium levels on salt diffusion; and (3) the feasibility of repurposing salty whey as an alternative salt source in cheese manufacturing.

MATERIALS AND METHODS

Preparation of MCC

To investigate salt diffusion using salty whey as a salting medium, a model system was prepared from MCC. The production protocol was adapted from that of Ale-

hosseini et al. (2021). Briefly, skim milk was pasteurized using a plate heat exchanger at 72°C for 15 s and subjected to microfiltration using a 1.40- μ m ceramic membrane (Tami Isoflux, Tami Industries, Nyons, France). The permeate was then passed through a secondary microfiltration membrane (0.14 μ m pore size, 0.35 m² surface area) at 50°C, and the final retentate was stored at –8°C until further use. The initial protein concentration of the MCC solution was 10% (wt/wt). To further concentrate the protein, samples were concentrated using rotary evaporation (Rotavapor Buchi R-300, Switzerland) at 45°C, 50 rpm, and 50 kPa vacuum for 20 to 30 min, increasing the protein content to 20% (wt/wt). Rotary evaporation under mild thermal conditions was applied due to its suitability and feasibility for laboratory-scale concentration. Should scale up be required, the use of a ceramic membrane microfiltration system may be considered. The protein concentrations were confirmed using the Kjeldahl method (IDF20B:1993; ISO, 1993).

Experimental Design

The experiment was designed to evaluate the effect of fat content, calcium concentration, pH, and salting temperature on salt diffusion in MCC gels using salty whey. Selected treatments were also compared with diffusion behavior in standard brine. The pH of the brine was adjusted to match that of salty whey (pH 5.59) by adding 0.1 M HCl, and calcium concentration was also adjusted to match that of salty whey by adding CaCl₂ to the brine.

The treatment variables included calcium chloride (CaCl₂, Sigma-Aldrich, Germany) concentrations of 0%, 0.5%, or 1% (wt/wt) and glucono- δ -lactone (GDL; C₆H₁₀O₆, Jungbunzlauer, Basel, Switzerland) at 0%, 1%, or 2% (wt/wt). In addition, cream separated from raw milk with ~40% fat content was blended with liquid MCC to give protein-to-fat ratios of 1:0.0, 1:0.28, or 1:0.66; the cream and MCC solution were heated separately up to 50°C and subsequently mixed thoroughly to obtain a uniform mixture.

Rennet-induced gelation was achieved by adding 40 μ L/mL of diluted chymosin (CHY-MAX Plus, Chr. Hansen, Denmark; 10% wt/wt, 200 international milk clotting units), followed by a 30-min holding time at room temperature. Gels were molded in 30-mL polystyrene centrifuge tubes (base removed), yielding cylindrical samples with diameter and height of 25 mm. Samples were then salted at 20°C, 30°C, or 40°C. Salting temperatures were selected based on commercial Cheddar salting and reports on Cheddar cheese salting conditions (Fox et al., 2017; Ong et al., 2020). A full factorial design (i.e., 3 levels $\times$ 3 variables $\times$ 3 temperatures) resulted in 81 treatment combinations for salt analysis, as reported in Supplemental Table S1 (see Notes).

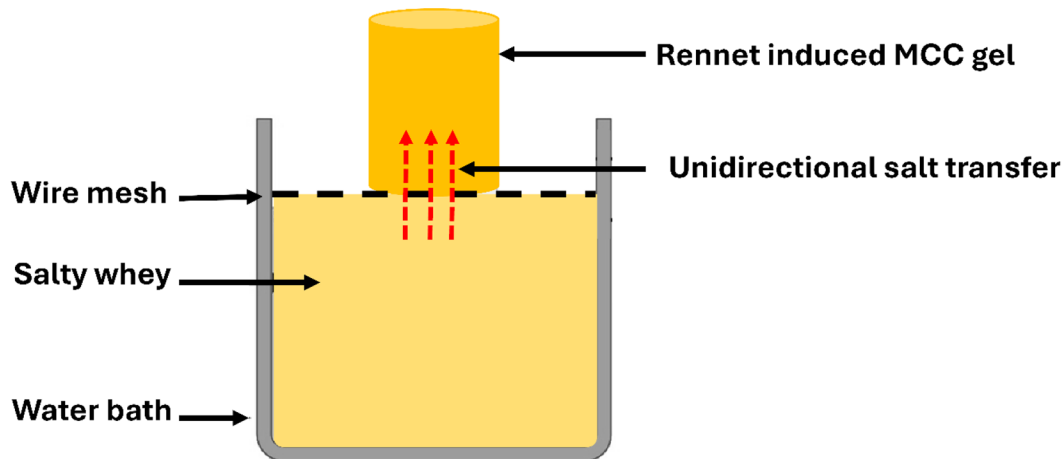


Figure 1. Schematic illustration of the salting process applied to rennet-induced MCC gel samples.

Salting Procedure

The salting setup is illustrated in Figure 1. Gel samples were placed in wire mesh holders to facilitate uniform, unidirectional solute transfer. The entire assembly was immersed in a temperature-controlled water bath (Sci-Quip, UK), with the salty whey (8.6% wt/wt NaCl) maintained under a constant slow agitation for 1 h. A 1:200 sample/salting medium ratio (2,000 mL salty whey) was maintained throughout.

Out of 81 treatment combinations, several selected treatments—including those determined to have the highest (ST-1), lowest (ST-2), and moderate (ST-3) levels of salt diffusion, alongside a control (0% CaCl_2 , GDL, fat) sample (ST-4), and a high-fat sample (ST-5)—were separately salted using brine to compare the salt diffusion in 2 salting mediums. The formulated brine replicated salty whey in terms of pH (5.59), NaCl content (8.6% wt/wt), and calcium content (79.2 mg/100 g). These samples were further analyzed for texture by texture analysis, functional group analysis by Fourier transform infrared (FTIR) spectroscopy, and microstructure using scanning electron microscopy and confocal laser scanning microscopy for validation of salt diffusion outcomes.

Measurement of Salt Content

Following salting, 1-mm thick slices were cut precisely 10 mm from the gel–salting medium interface using a single-edge blade (009-rd, GEM). Sodium chloride content was determined in MCC gels via chloride ion quantification using the potentiometric titration method (IDF88:2006; ISO, 2006). For formulations containing added CaCl_2 , corresponding chloride content was subtracted from the final measurement. Based on the process

described in IDF229:2018 (ISO, 2018) for mineral analysis, inductively coupled plasma optical emission spectroscopy (ICP-OES) was used to determine the added chloride content in MCC gels (Agilent 5110, vertical dual view ICP-OES analyzer), following the digestion of 0.25 g of MCC gel in 69% concentrated HNO_3 at 210°C using a microwave digester (CEM Microwave Technology), as described in established protocols for chloride quantification in dairy matrices (Naozuka et al., 2003).

Diffusion Coefficient Calculation

The diffusion coefficient of salt was calculated from the above 1-mm thick slices of MCC gel samples using Equation 5. Several assumptions were made for the determination of the salt diffusion coefficient in MCC gels. The system was considered a defined unit with stable boundaries, and solutes (i.e., salt molecules) were assumed to migrate unidirectionally along the axial length of the cylindrical MCC gel samples (Cu and Saltzman, 2009). Throughout the study, the samples were exposed to a constant concentration of brine or salty whey. Volumetric changes such as extensive swelling, which are often correlated with non-Fickian diffusion behavior, were considered negligible (Crank, 1979). Additionally, given the high ratio of sample to brine or salty whey volume, mineral transfer from the gel samples to the brine was assumed to be negligible and thus omitted from the calculations.

Fick's first law, applicable only under steady-state conditions, was employed to model the system, where the molar flux is proportional to the concentration gradient of solutes and the rate of concentration change with respect to distance (Floury et al., 2010). Equation 1 shows Fick's first law,

$$J = D \frac{\partial C}{\partial x}, \quad [1]$$

where J is the diffusion flux or total rate of solute transfer per area ($\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$), D is the diffusion coefficient ($\text{m}^2 \cdot \text{s}^{-1}$), C is the solute concentration (mol/L), and x is the cross section of the direction of diffusion (m).

In contrast, for non-steady-state conditions where the concentration of the solutes changes over the time (t) at a given spatial distance (d), Fick's second law can be applied as follows (Crank, 1979):

$$\frac{dCx}{dt} = \frac{d}{dx} \left(D \frac{dCx}{dx} \right). \quad [2]$$

When D is not equal to Dx , this equation can be represented as follows:

$$D \neq Dx:$$

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}. \quad [3]$$

- For $t = 0$, $C = C_0$ at $0 \leq x \leq \infty$.
- For $t > 0$, $C = C_s$ (surface concentration) at $x = 0$.

$$C = Cx \text{ at } x = \infty.$$

$$C(x, t) - C_0 = \frac{C_s - C_0}{\sqrt{\pi}} \int_{-\infty}^{\frac{x}{2\sqrt{Dt}}} e^{-\xi^2} d\xi = \frac{C_s - C_0}{2} \left[1 = \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right) \right]. \quad [4]$$

Then,

$$\frac{Cx - C_0}{C_s - C_0} = 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right), \quad [5]$$

where C_0 refers to the initial solute concentration in sample, solute concentration of a particular distance after solute migration from the interface is denoted by Cx , and the surface concentration of solutes in brine or salty whey diffusing into the surface is denoted by C_s . C_0 , Cx , and C_s are measured in mol/L . The error function (erf) is defined as follows:

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-\xi^2} d\xi. \quad [6]$$

Physicochemical Properties

The physicochemical properties of the samples were analyzed using standard methods. Moisture content and TS were determined by oven drying (IDF4:2004; ISO, 2004), protein content was measured using the Kjeldahl method (IDF20B:1993; ISO, 1993), and ICP-OES (Agilent 5110, vertical dual view ICP-OES analyzer) was used for quantification of calcium concentration (IDF229:2018; ISO, 2018). The pH of the samples was recorded using a digital pH meter (HQ11d, Hach), which was calibrated before each measurement.

Texture Analysis

Texture properties of the selected rennet-induced MCC gels were evaluated using a texture profile analyzer (TA.XTplus, Stable Micro Systems, UK). Both compression and penetration tests were conducted on cylindrical samples (25 mm diameter $\times$ 25 mm height) using a 5.00-kg load cell. The compression test was performed by applying a 50% strain with a compression plate probe (type: P/75) at pretest, test, and post-test speeds of 2, 1, and 2 mm/s, respectively. For the penetration test, force was applied perpendicular to the center of the cylindrical sample to a depth of 5 mm using a stainless steel probe (type: P/6) with a trigger force of 0.01 N at 25°C. The pretest, test, and post-test speeds were identical to those used in the compression test. Textural parameters were calculated using Texture Exponent Software (v.6.1.18.0, Stable Micro Systems, UK).

Microstructural Analysis

Scanning electron microscopy was used for investigate the effects of variations in pH, calcium, and fat concentrations on the microstructural properties of rennet-induced MCC gels (Zeiss Supra 40VP, Germany), following the method of Alehosseini et al. (2021) with slight modifications. Before imaging, samples were freeze-dried using a benchtop freeze-dryer (FreeZone 6L, Labconco) and subsequently sputter-coated with a gold-platinum mixture (Emitech K575X, UK). Final images were captured at an acceleration voltage of 2 kV. Scanning electron microscopy images were analyzed by ImageJ (Java 1.8.0_345) software developed by the National Institutes of Health; the width of the protein strands was estimated from the average of 10 different data points. The porosity of the gels was calculated as a ratio between area of pores and total image area and expressed as a percentage value.

Confocal laser scanning microscopy was used to evaluate the distribution of protein and fat fractions within

Table 1. Composition and pH of the MCC gel control sample (0% GDL, 0% CaCl₂, 0% fat) and the salty whey used¹

Sample	Moisture (%)	Total solids (%)	Protein (%)	Ash (%)	Ca (mg/100 g)	pH	Salt % (wt/wt)
MCC gel	75.49 ± 1.09	24.15 ± 1.19	20.19 ± 0.31	5.61 ± 0.25	5,719 ± 139	6.45 ± 0.02	NR ²
Salty whey	91.53 ± 0.03	8.35 ± 0.36	0.89 ± 0.09	7.98 ± 0.18	79.2 ± 8.3	5.59 ± 0.01	8.6 ± 0.1

¹Values are presented as mean ± SD.²NR = not relevant.

the gel matrix (Leica TCS SP5, Leica Microsystems, Germany), as described by Alehosseini et al. (2023) with some modifications. The MCC gels were sliced into 2 mm (height) × 25 mm (diameter) sections using a single-edge blade and mounted on microscope slides (Fisher Scientific). A 20-μL aliquot of each dye solution (i.e., the protein-specific dye [Fast Green, Sigma-Aldrich; 0.01% wt/vol] and the fat-specific dye [Nile Red]), was applied to the sample surface. Excitation wavelengths of 633 nm (He-Ne laser) for Fast Green and 488 nm (Ar laser) for Nile Red were used, with images collected at an emission wavelength range of 650 to 700 nm.

Attenuated Total Reflection FTIR Analysis

The structural characteristics of freeze-dried MCC gels were assessed using FTIR spectroscopy (Invenio INV-0013041519, Bruker, Germany) in attenuated total reflectance mode over a spectral range of 650 to 4,000 cm⁻¹ with OPUS software (version 8.7.41; Bruker Optics GmbH). Before analysis, the performance qualification test and background signal channels were tested for any deviations. Each spectrum was obtained as an average of 50 scans. Atmosphere compensation, H₂O and CO₂ compensation, and vector normalization were carried out before spectral validation (Fellah et al., 2009).

Statistical Analysis

Data were analyzed by one-way ANOVA using IBM SPSS Statistics software (version 27, IBM Corp.). Duncan's multiple range test was applied to compare mean values, with differences considered significant at $\alpha = 0.05$. Salt analysis for selected treatments was conducted in duplicate, whereas all other analyses were performed in triplicate.

RESULTS AND DISCUSSION

Physicochemical Properties of Salty Whey and MCC Gel

The TS content of the salty whey used in this study falls within the ranges reported by Blaschek et al. (2007), who observed TS content of 6.1% to 24.1% (wt/wt) for

salty whey derived from commercial Cheddar cheese production. However, the moisture content of the MCC gel used in the present study was considerably higher than typical cheese matrices at the salting stage (Table 1). For instance, the moisture content of Cheddar curd ranges between ~40% (wt/wt; Ong et al., 2020) and 43% to 47% (wt/wt; observed during pilot scale Cheddar cheese production in this research center) just before milling. The MCC gel in this study was formulated to contain ~20% (wt/wt) protein (Table 1). Further concentration of the protein content in the MCC by rotary evaporation was not feasible due to excessive viscosity and poor flowability of the suspension as it became more concentrated. In contrast, Cheddar cheese typically contains 22.7% to 28.5% (wt/wt) protein (Hickey et al., 2013).

Variations in calcium content observed in the present study may be attributed to differences in the source of salty whey, particularly variations in the salting method, salting duration, and initial salt application rates. These values are consistent with previous reports, though the absolute values vary depending on whey processing and cheese types. Alehosseini et al. (2021) reported calcium concentrations of ~500 mg/100 g in MCC and 750 mg/100 g in Cheddar curd, which are consistent with the values observed in the present study. Talebi et al. (2019) found calcium concentrations of 130 ± 8 mg/100 g in salty whey permeate obtained in dairy plants from Victoria, Australia, following ultrafiltration to remove the whey protein, whereas Nishanthi et al. (2017) reported a calcium concentration of 72 ± 0.7 mg/100 g from salty whey obtained from the same region.

Similarly, a compositional discrepancy in ash content (milk minerals plus added salt) of the salty whey was noted in current study. Nishanthi et al. (2017) reported as ash contents as high as 14.8% (wt/wt) in salty whey, nearly double the levels observed in present study, whereas Al Kader et al. (2023) found that the ash content of salty whey produced from full-fat Cheddar cheese was ~8.92% (wt/wt). Such variability is likely due to differences in the composition of the cheese matrix, salt inclusion levels, and processing parameters across different production facilities.

The pH of the salty whey used as the salting medium in this study was 5.59 (Table 1), which is slightly higher than the pH range typically observed in brine formed at

the curd chip surface during Cheddar cheese salting (approximately pH 5.0–5.3; Guinee and Fox, 2017). This pH difference may also influence the salt diffusion behavior in the MCC gels, underscoring the importance of standardizing pH conditions when comparing salting systems.

Factors Influencing Salt Diffusion in MCC Gels

In the current study, the NaCl concentration of salty whey was 8.6% (wt/wt), which lies within the range where salt concentration has a minimal influence on diffusion. A salt concentration between 5% and 20% in brine has a limited influence on rate of salt diffusion (Guinee, 2004); however, salt uptake increases at diminishing rate when the concentration exceeds 25% (wt/wt; Melilli et al., 2003). Nevertheless, the solubility of NaCl in water is limited to ~26.4% (wt/wt) at 25°C (Vane, 2017). Therefore, it is unlikely that the initial NaCl content of the salty whey substantially contributed to differences in salt absorption among the MCC gel samples.

The salt diffusion coefficients (D) of the rennet-coagulated MCC gels in this study, with pH values ranging from 6.63 to 5.13, ranged from 3.1×10^{-9} m²/s to 8.5×10^{-9} m²/s which are substantially higher than those reported by Turhan and Kaletunc (1992), who observed D values ranging from 0.20 to 0.39×10^{-9} m²/s in white cheese at pH 5.3 brined at 4°C to 20°C using brine at 15% to 20% (wt/wt) salt.

Similarly, in our earlier study involving MCC gels with a maximum protein content of 15% (wt/wt), the highest

recorded D value under brining conditions was 0.86×10^{-9} m²/s (Alehosseini et al., 2023). The substantially higher diffusion rates observed in the present study are likely attributable to the combined effects of increased protein concentration (20% wt/wt), elevated calcium levels (up to 1% wt/wt, which gave levels of $1,478.7 \pm 25.5$ mg/100 g in the final model cheese), and optimized salting temperature (30°C), all of which promote enhanced salt transport. The structure of the MCC gel matrix is integral to salt diffusion behavior. The casein network is known to exert a sieving effect, regulating both the inward diffusion of Na⁺ and Cl⁻ ions and the outward migration of water (Guinee and Fox, 2017).

Geurts et al. (1974) studied the nonuniformity of pore size and orientation within protein matrices and emphasized how such heterogeneity can restrict or facilitate solute migration depending on microstructural features. In lower-protein gels, planar regions near the brine interface may develop with minimal solute exchange, increasing the effective diffusion path. In contrast, higher protein concentrations elevate the protein-to-fat ratio, which in turn raises moisture retention (Geurts et al., 1981; Giroux et al., 2022). These mechanistic insights are consistent with the elevated D values observed in this study at high CaCl₂ inclusion levels for the MCC gel treatments (Table 2).

Among the tested variables, calcium content and salting temperature had the most significant effects ($P < 0.05$) on salt diffusion (Table 3, Table 4, and Figure 2). McMahon et al. (2005) reported that the addition of

Table 2. Sodium chloride concentrations and calculated salt diffusion coefficients (D) for selected MCC gel treatments as a function of GDL, CaCl₂, fat addition levels at formulation, and salting temperature¹

S/N	ST/NS	GDL (%)	CaCl ₂ (%)	Fat (%)	Temperature (°C)	pH	NaCl (%)	D (m ² /s)
1	ST-2	1	0	10	20	5.91	0.293	3.09×10^{-9L}
2	ST-5	0	0	10	20	6.51	0.401	3.51×10^{-9K}
3	NS	0	0	10	30	6.34	0.421	3.58×10^{-9J}
4	NS	1	0	5	30	6.03	0.462	3.73×10^{-9I}
5	NS	1	0	5	20	6.01	0.533	3.99×10^{-9H}
6	NS	0	0	5	40	6.63	0.539	4.01×10^{-9H}
7	ST-4	0	0	0	30	6.40	0.540	4.01×10^{-9H}
8	NS	0	0.5	5	30	6.26	0.870	5.17×10^{-9G}
9	ST-3	2	0.5	5	20	5.50	0.875	5.19×10^{-9G}
10	NS	2	0	5	30	5.54	0.880	5.20×10^{-9FG}
11	NS	0	0	5	20	6.55	0.893	5.25×10^{-9EF}
12	NS	0	0.5	5	40	6.32	0.906	5.30×10^{-9DE}
13	NS	0	0.5	10	40	6.29	0.913	5.32×10^{-9D}
14	NS	1	1	0	30	5.49	1.40	7.10×10^{-9C}
15	NS	1	1	10	20	5.6	1.53	7.63×10^{-9B}
16	ST-1	0	1	0	30	6.03	1.74	8.53×10^{-9A}

^{A-L}Means within a column with different superscripts are significantly different ($\alpha = 0.05$).

¹S/N denotes the serial number, and NS denotes as not selected for further analysis. ST refers the corresponding selected treatment number for the further analysis, including validation of salt diffusion experiment and comparison of salty whey with brine. ST-1 refers to the selected treatment having highest salt migration; ST-2 refers to the treatment having lowest salt migration; ST-3 refers to the treatment having moderate salt migration; ST-4 refers to the control treatment; and ST-5 refers to the treatment with the highest fat included.

Table 3. Statistical summary of factors affecting salt diffusion rates in MCC gel samples

Factor	<i>P</i> -value ¹
GDL (%)	0.183
CaCl ₂ (%)	0.035*
Fat (%)	0.078
Temperature (°C)	0.043*
GDL × CaCl ₂	0.324
GDL × Temperature	0.354
CaCl ₂ × Temperature	0.901

¹No interaction effects were significant.**P* < 0.05.

calcium chloride at higher levels (e.g., 0.6% wt/wt) improved rennet gel homogeneity by increasing the number of serum pockets and promoting protein aggregation into larger structures. In addition, calcium can reduce syneresis at specific pH levels, thereby helping to retain moisture and enhancing the accessibility of diffusion pathways (Panthi et al., 2019). However, the presence of calcium in the salting medium can exert the opposite effect; Luo et al. (2013) reported that Mozzarella cheese salted with high-calcium brines (0.1%–0.25% wt/wt) absorbed less NaCl, likely due to a denser protein matrix and reduced interstitial space, both of which hinder ion mobility.

The influence of pH was modulated through the addition of GDL, a slow-releasing acidulant. Increasing the GDL addition level from 0% to 2% (wt/wt) reduced the pH of MCC gels from 6.63 to 5.13. Salt diffusion initially declined as pH decreased from 6.6 to 6.0 but then increased as pH continued to decrease. This biphasic trend may be attributed to the interplay between matrix porosity and syneresis. At lower pH, the casein matrix becomes more porous, facilitating salt entry, while also reducing water loss during brining, potentially due to pH buffering and osmotic pressure modulation with increased lactate concentrations (Kemp, 2005; Alehosseini et al., 2023). However, the reduction in salt uptake between pH 6.6

and 6.0 may be linked to pH-induced syneresis, which tightens the matrix and limits solute transport (Fagan et al., 2017). Similar observations were made by Geurts et al. (1981), who reported reduced salt uptake in Gouda cheese as pH increased from 4.7 to 5.7, at similar moisture contents.

Salting temperature also played a critical role in determining diffusion behavior. In Cheddar-like hard cheese, salting temperature is typically ~38°C (Ong et al., 2020). Exceeding this temperature would result in melting most of the fat, and its subsequent presence at the cheese surface. This phenomenon was evident as a reduced amount of fat present in the curd surface at 26°C compared with 32°C (Sutherland, 1974; Ong et al., 2017). In contrast, the present study identified 30°C as the optimum temperature for salt migration, with significantly (*P* < 0.05) lower diffusion observed at 20°C or 40°C. Prior studies have consistently shown that increasing brining temperature from 5°C to 20°C enhances both salt uptake and diffusion rates (Geurts et al., 1974; Guinee and Fox, 1986; Turhan and Kaletunc, 1992). In contrast, a decrease in the curd temperature at salting increases the salt in moisture (S/M) of the final cheese during dry salting of Cheddar cheese (Ong et al., 2017); cheese showed a reduction of salt and S/M as the temperature of curd increased from 24°C to 41°C. This reduction may be due to salt loss caused by whey expulsion from Cheddar curd chips at high temperature (Fox et al., 2017).

At lower temperatures (e.g., 12°C), increased aqueous phase viscosity may impede the expulsion of serum, thereby hindering salt penetration (Guo et al., 1997). Although our results align with these findings, contrasting evidence from Melilli et al. (2003) indicated that lower temperature (18°C) and lower brine concentration (18% wt/wt) could actually enhance salt uptake due to the high porosity of the cheese near to the block surface and reduced moisture loss. At higher salting temperatures (e.g., 40°C), salt diffusion may be compromised by shrinkage of the casein matrix. Geurts et al. (1981)

Table 4. Formulation and final pH and moisture content of selected MCC gels

Sample formulation				Properties of the final MCC gel samples		
Selected treatment	GDL% (wt/wt)	Ca% (wt/wt)	Fat% (wt/wt)	pH	Moisture content (%)	Remarks/ <i>D</i> value ¹ (m ² /s)
ST-1	0	1	0	6.03 ± 0.17 ^B	76.24 ± 1.18 ^{AB}	Highest salt migration (8.53 × 10 ⁻⁹) ^A
ST-2	1	0	10	5.91 ± 0.14 ^B	73.96 ± 1.81 ^{AB}	Lowest salt migration (3.09 × 10 ⁻⁹) ^L
ST-3	2	0.5	5	5.50 ± 0.15 ^C	72.42 ± 1.9 ^B	Moderate salt migration (5.19 × 10 ⁻⁹) ^G
ST-4	0	0	0	6.40 ± 0.10 ^A	77.59 ± 0.81 ^A	Control sample (4.01 × 10 ⁻⁹) ^H
ST-5	0	0	10	6.51 ± 0.06 ^A	76.61 ± 1.92 ^{AB}	Highest fat sample (3.51 × 10 ⁻⁹) ^K

^{A–L}Means ± SD within a column with different superscripts are significantly different (α = 0.05).¹Salt migration outcomes are based on NaCl percentage at 1 cm height at 1 h duration and respective coefficient of diffusion value (*D* value).

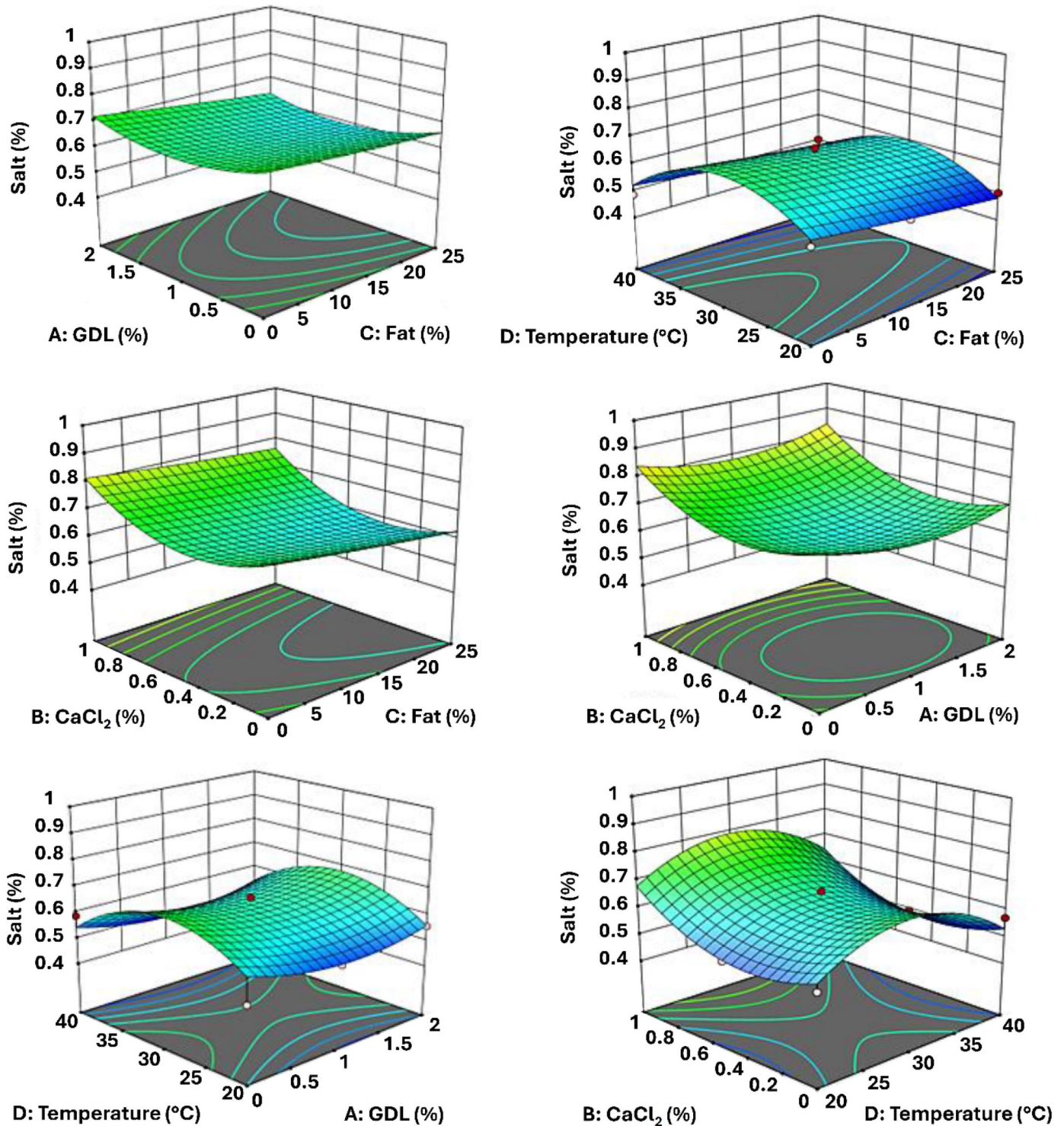


Figure 2. Surface plots of salt concentration in rennet gels at 1 cm height as affected by the percentage of GDL, CaCl₂, and fat in the model cheese formulation at differing salting temperatures; all percentages expressed in wt/wt basis.

noted that elevated temperatures could promote matrix compaction due to enhanced hydrophobic interactions among proteins, which increase system free energy. This

shrinkage may partially counterbalance osmotic driving forces, thereby limiting the net migration of salt (Fucà et al., 2012).

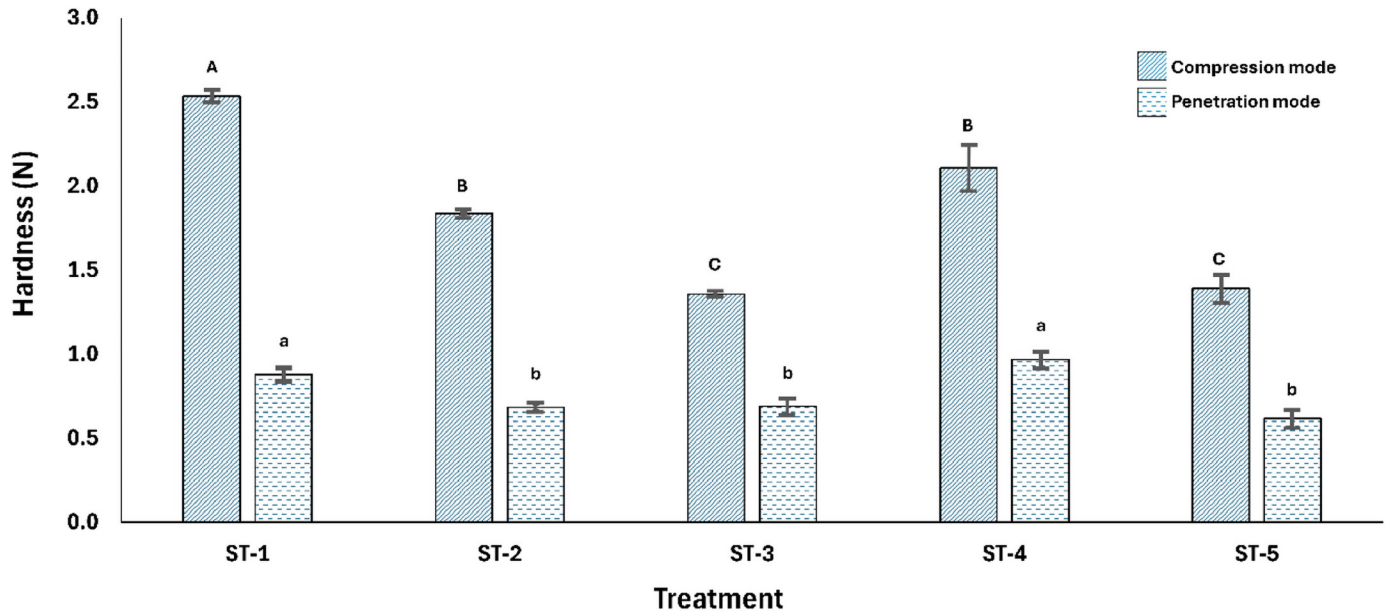


Figure 3. Values for hardness in compression mode and penetration mode for selected MCC gel samples (ST-1 = highest salt diffusion, ST-2 = lowest salt diffusion, ST-3 = moderate salt diffusion, ST-4 = control, ST-5 = high-fat sample). Values are expressed as mean $\pm$ SD for 3 replicates. Values with different uppercase letters (A–C) are significantly different ($P < 0.05$) in hardness in compression mode, and values with different lowercase letters (a–c) are significantly different ($P < 0.05$) between samples for hardness in penetration mode.

Regarding the effect of fat content, the MCC gel sample with the highest level of fat inclusion at formulation (10% wt/wt) exhibited the lowest salt diffusion. Although the effect of fat content was not significant ($P = 0.078$) across all treatments, a general trend of decreased diffusion with increasing fat levels was evident (Tables 2 and 4). Isolating the effects of individual components is challenging, as compositional changes often affect multiple parameters simultaneously. However, the fat-to-SNF and moisture-to-SNF ratios are known to influence salt diffusion coefficients by altering the distribution and tortuosity (indirect routes due to obstructions in the matrix that salt ions have to pass through as they move in the cheese matrix) of protein and fat domains within the matrix (Guinee, 2004).

Geurts et al. (1974) reported curvilinear increases in D values in cheese samples as protein and fat content decreased in cheese samples with identical fat-in-DM. Although higher fat levels can obstruct diffusion pathways, forcing salt ions to travel around fat globules, the accompanying reduction in protein content may mitigate this effect. In fact, cheeses with lower fat levels typically exhibit higher D values due to reduced obstruction (Geurts et al., 1974). However, in practical scenarios, reducing fat leads to higher moisture and protein concentrations, potentially increasing matrix tortuosity and lowering D . For example, in Gouda cheese with a constant moisture level (50% wt/wt), increasing fat content from 11% to 26% (wt/wt) increased D values from 0.15 to

0.25 cm^2/d (Geurts et al., 1974), likely due to concurrent decreases in protein matrix density. Conversely, Estrada et al. (2019) reported that raw sheep milk cheeses with higher fat content (55.6% DM) exhibited high salt concentrations compared with low fat content (52.9% DM), highlighting the complexity and product-specific nature of these interactions.

Structural and Microstructural Contributions to Salt Diffusion

Correlation Between Salt Diffusion and Texture Properties. Texture profile analysis was performed on selected rennet-induced MCC gel samples to assess the relationship between mechanical properties and salt diffusion behavior. The sample exhibiting the highest salt diffusion (ST-1) required the greatest force for compression, followed by the control sample (ST-4). In contrast, the samples with the lowest salt diffusion (ST-2 and ST-5) required the least penetration force ($P < 0.05$; Figure 3).

In the present study, the highest hardness value of 2.53 ± 0.8 N was observed in sample ST-1, which contained highest calcium levels at formulation (1% wt/wt, corresponding to $1,478.7 \pm 25.5$ mg/100 g in final model cheese). This resulted in a significant increase in compression force ($P < 0.05$) compared with the other selected treatments. The results also suggest that increased calcium content enhances matrix hardness, aligning with previous findings in Swiss cheese, where a high level of

calcium phosphate contributed to a more firmer protein matrix (Mandl et al., 2009). Santos et al. (2013) similarly reported that calcium addition improved textural firmness in Minas cured cheese, irrespective of pH.

Hardness, measured as the maximum force during compression, was highest in the ST-1 sample, which contained the greatest concentration of calcium chloride. This observation is consistent with the principle that added divalent cations increase ionic strength, promoting protein solubility and swelling, which in turn enhance protein–water interactions (Pastorino et al., 2003). The resulting expansion of protein strands improves the mechanical integrity of the matrix and its ability to resist deformation (Paulson et al., 1998). However, Soodam et al. (2015) reported no significant effect of added calcium on the texture of Cheddar cheese, suggesting potential variability depending on product type or formulation.

Adhesiveness, defined as the work required to overcome attractive forces between the cheese and the contacting material, was highest in the ST-5 sample (10% fat), followed by ST-2 (also with 10% fat). Similar trends have been reported in Cheddar cheeses, where samples with higher fat levels (27%–34% wt/wt) showed greater adhesiveness compared with lower-fat counterparts (13%–21% wt/wt; Bryant et al., 1995).

Beyond structural effects, factors that influence protein–water and protein–protein interactions may also affect adhesiveness. Pastorino et al. (2003) reported that NaCl injection into unsalted Muenster cheese increased adhesiveness by strengthening the existing protein interactions with water and other nonprotein components. These observations are in good agreement with the current findings.

Microstructural Analysis of Samples. Salt diffusivity is not only determined by the physicochemical properties of the solutes, but also by the compositional and microstructural characteristics of the protein matrix (Floury et al., 2010). Therefore, microstructure analysis was conducted to elucidate the effect of fat content, pH, and calcium concentration on the internal architecture of rennet-induced MCC gels.

As illustrated in Figure 4, the most salt-permeable sample (ST-1) exhibited a dense and thick protein network, whereas the control sample (ST-4) displayed a looser, thinner matrix. Higher calcium content (1% wt/wt) clearly enhanced the wall thickness and rigidity of the protein network in ST-1 (Figure 4 and Table 5). These changes are consistent with previous reports indicating that calcium promotes the formation of structural features such as dead-end channels which influence solute movement (Alehosseini et al., 2023). Interestingly, such channels were also more frequently observed in the control sample (ST-4), which had the highest porosity ($43.96\% \pm 1.11\%$, Table 5), suggesting that calcium may

reduce excessive porosity while also enhancing overall network cohesion.

A less porous architecture was observed in sample ST-2 (lowest salt diffusion; Table 5) compared with other treatments ($P < 0.05$), suggesting that a porous matrix can facilitate salt transport. Although pore size (diameter) is a key determinant of diffusion behavior, it could not be conclusively quantified from the current micrographs. Nevertheless, the distribution of fat also played a clear role in shaping the protein matrix. In samples ST-2 and ST-5, high fat content (10% wt/wt) led to disrupted protein bonding and heterogeneous matrix structures, reducing compactness and rigidity. Additionally, fat appeared to coat the protein network, which may obstruct salt penetration.

These microstructural observations closely align with the measured salt diffusion behavior. Samples with more continuous and compact matrices permitted greater inward salt migration with limited water loss, whereas looser or fat-coated matrices hindered diffusion. Thus, morphological parameters (i.e., network density, wall thickness, porosity, and fat distribution) substantially influenced diffusion dynamics in MCC gels prepared under different compositional regimes.

Confocal laser scanning microscopy provided additional insights, albeit at lower resolution for protein networks. However, fat globule distribution was clearly visualized (Figure 5), showing various sizes and degrees of aggregation across treatments. Curds with higher salt concentrations (15%–25% wt/wt) showed significantly altered fat distribution in the cheese matrix, with wide range of fat particle size (0.2 μm to 100 μm) probably caused by disruption and coalescence, whereas unsalted or low-salt curds had more uniform fat particles (Ong et al., 2020). The present study confirmed that fat was heterogeneously distributed and may have obstructed salt diffusion channels within the protein matrix.

As shown in Figure 5, sample ST-1 exhibited a denser protein network than the control sample (ST-4), further

Table 5. Width of protein strands and porosity (%) of gels as assessed from scanning electron microscopy images of selected MCC gel samples at a magnification level of $\times 100^1$

Selected treatment ²	Width of protein strands (nm)	Porosity (%)
ST-1	24.26 ± 5.8^A	31.56 ± 1.82^B
ST-2	15.47 ± 3.0^B	25.11 ± 2.21^C
ST-3	16.44 ± 2.8^B	33.62 ± 1.83^B
ST-4	12.49 ± 2.4^B	43.96 ± 1.11^A
ST-5	13.07 ± 1.6^B	28.96 ± 0.15^{BC}

^{A–C}Means within a column with different superscripts are significantly different ($\alpha = 0.05$).

¹Values are presented as mean $\pm$ SD.

²ST-1 = highest salt diffusion; ST-2 = lowest salt diffusion; ST-3 = moderate salt diffusion; ST-4 = control; ST-5 = high-fat sample.

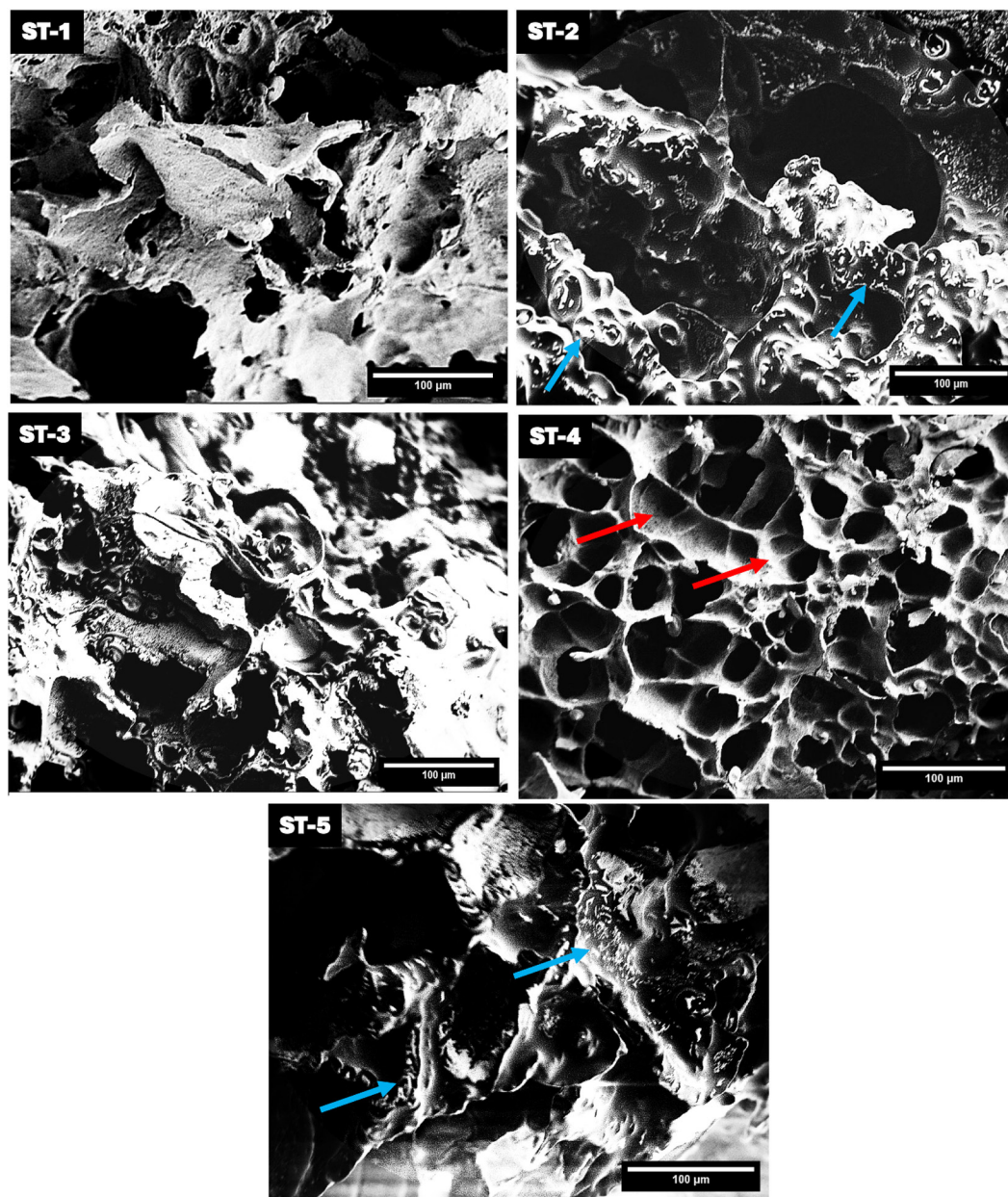


Figure 4. Scanning electron microscopy images of MCC gel samples: ST-1 = highest salt diffusion, ST-2 = lowest salt diffusion, ST-3 = moderate salt diffusion, ST-4 = control, ST-5 = high-fat sample. Red arrows indicate dead-end channels, and blue arrows highlight fat glazing on the protein matrix. All images are at the magnification level of 250 $\times$.

confirming the role of calcium in enhancing cross-linking and matrix cohesion. These structural modifications provide mechanistic support for the differences in salt diffusion measured across treatments.

Fourier Transform Infrared Analysis. Fourier transform infrared spectroscopy was used to analyze the composition of the functional groups in samples. As shown in Figure 6, all rennet-induced MCC gel samples

exhibited characteristic absorption peaks with varying intensities. The spectral region between 3,800 and 3,100 cm^{-1} corresponds to O–H stretching and N–H vibrations associated with hydroxyl groups and the amide A band of polypeptides and amino acids (Woodcock et al., 2008). The position and width of these bands are informative indicators of hydrogen bonding strength (Spasov et al., 2006). A peak centered near 3,270 cm^{-1} is typi-

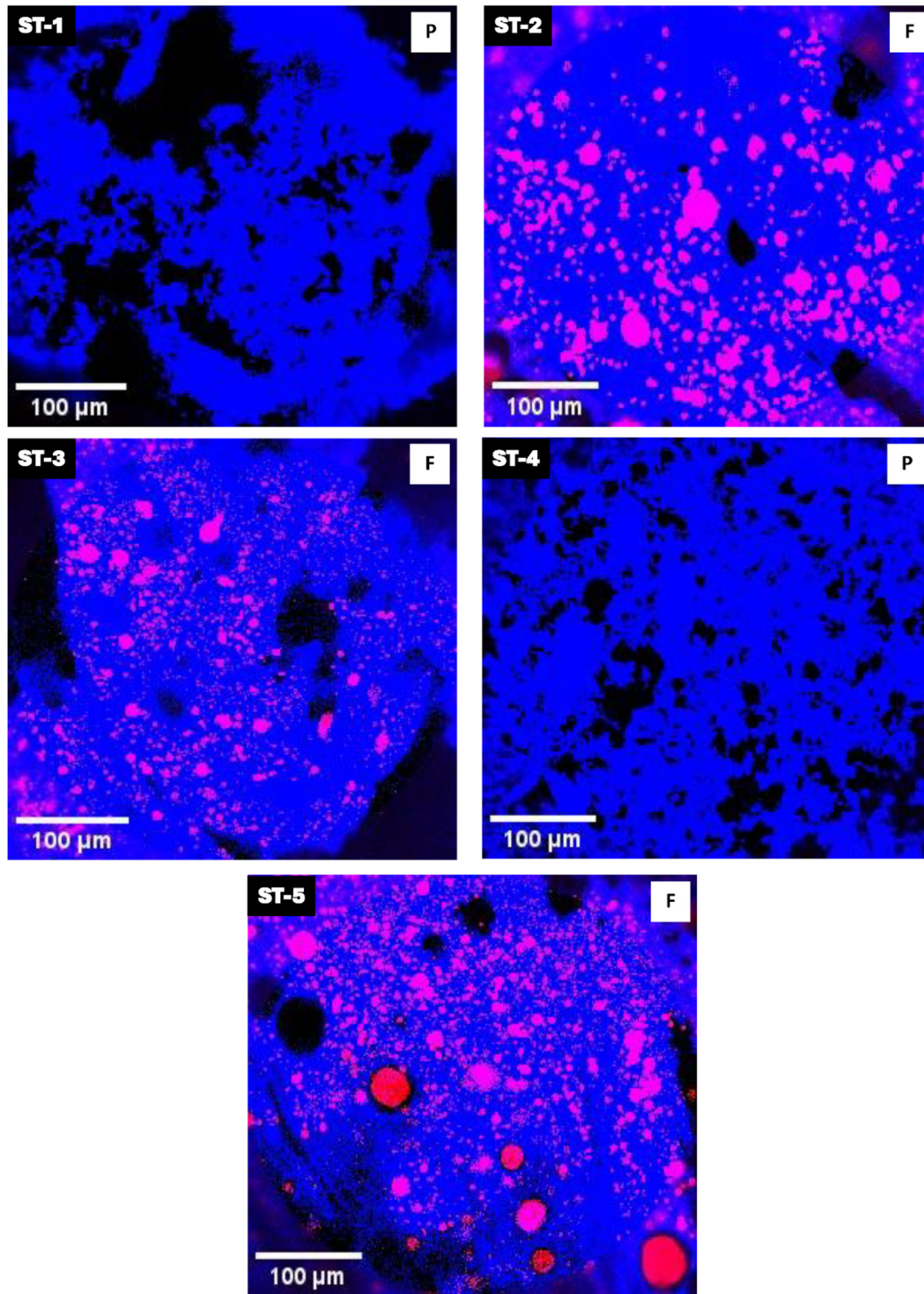


Figure 5. Confocal laser scanning microscopy images of MCC gel samples at $63\times$ magnification. ST-1 = highest salt diffusion, ST-2 = lowest salt diffusion, ST-3 = moderate salt diffusion, ST-4 = control, ST-5 = high-fat sample. P = protein matrix; F = entrapped fat.

cally associated with medium-strength hydrogen bonds characteristic of β -sheet structures (Di Foggia et al., 2011). Conversely, N–H vibrations occurring at slightly

higher wavenumbers ($3,275\text{--}3,286\text{ cm}^{-1}$) are indicative of weaker hydrogen bonding and are often attributed to predominant α -helical content.

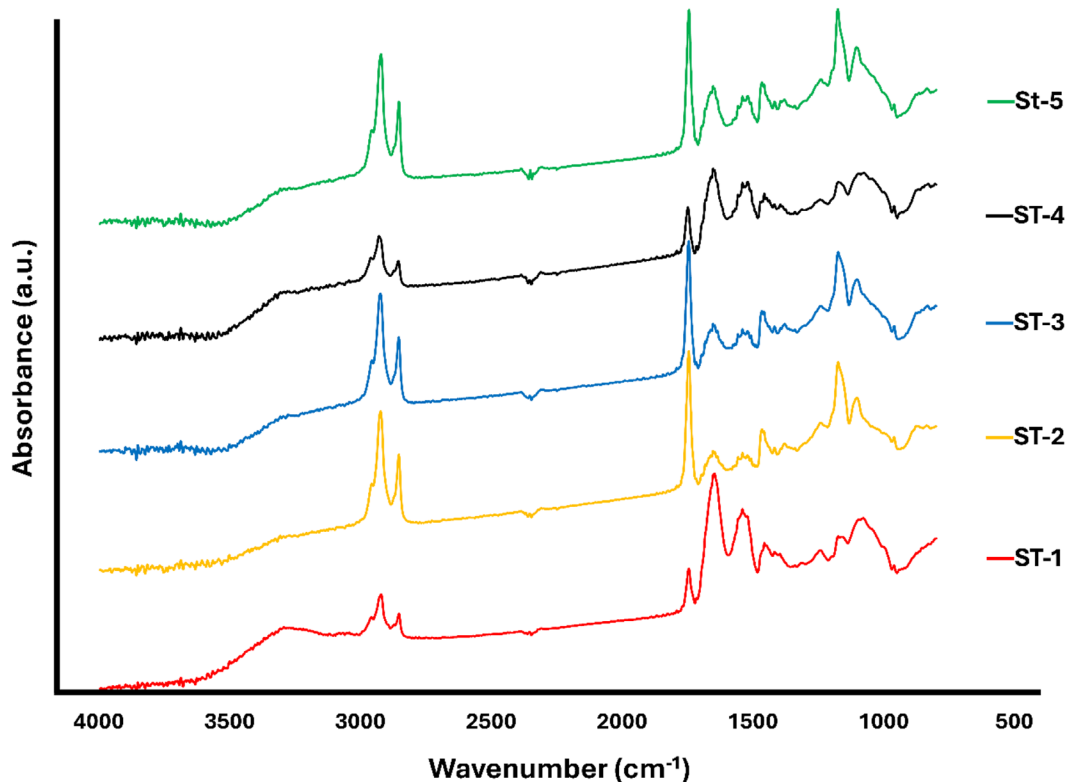


Figure 6. Fourier transform infrared spectra of rennet-induced MCC gel samples: (ST-1 to ST-5) salted using salty whey and brine (ST-1 = highest salt diffusion, ST-2 = lowest salt diffusion, ST-3 = moderate salt diffusion, ST-4 = control, ST-5 = high-fat sample). Key molecular vibrations reflect differences in protein secondary structure and lipid content. a.u. = absorbance units.

In the current study, treatment ST-1 (highest salt diffusion) exhibited a pronounced peak at $3,271\text{ cm}^{-1}$, indicating the presence of medium-strength hydrogen bonds. This suggests a more stable and structured protein matrix capable of retaining water and resisting syneresis (Tan et al., 2021). These findings align with the textural and microstructural properties described in the previous sections (Figures 3, 4, and 5).

The region spanning $3,100$ to $2,800\text{ cm}^{-1}$ encompasses symmetric and asymmetric C–H stretching vibrations, commonly associated with fatty acids (Foda et al., 2013). Samples with higher fat content (ST-2 and ST-5) showed strong absorption around $2,920\text{ cm}^{-1}$, confirming the presence of lipid-rich domains.

The amide I region ($1,630$ – $1,695\text{ cm}^{-1}$), primarily due to C=O stretching vibrations, provides insight into protein secondary structures. Treatment ST-1 displayed an intense peak at $1,642\text{ cm}^{-1}$, indicative of extended β -sheet structures stabilized by robust hydrogen bonding. Additionally, the amide II region ($1,510$ – $1,560\text{ cm}^{-1}$), associated with N–H bending and C–N stretching, showed a prominent peak at $1,539\text{ cm}^{-1}$ in ST-1, which corresponds to α -helical content (DeFlores et al., 2009). The high intensity of peaks in both amide I and II re-

gions in ST-1 confirms the presence of well-ordered and stable protein structures, likely contributing to enhanced salt migration through increased matrix coherence and reduced moisture loss. Collectively, the FTIR data reinforce previous findings from texture and microscopy analyses, demonstrating that samples with higher salt diffusion are characterized by more stable secondary structures and stronger hydrogen bonding networks.

Effects of Salting Media: Brine Versus Salty Whey

The salt migration in MCC gel samples exposed to either traditional brine or salty whey under similar conditions is shown in Figure 7. To determine the effects of each salting medium, critical parameters including pH (5.59), NaCl concentration (8.6% wt/wt), and calcium content (79.2 mg/100 g) were adjusted to equivalent levels in both solutions.

Despite these controls, treatments ST-1 and ST-5 displayed significant differences in the extent of salt diffusion between the 2 salting media (Figure 7). Brine consistently gave higher salt migration, and salty whey was less effective across nearly all samples ($P < 0.05$; in ST-1 and ST-5). This discrepancy is likely attributable

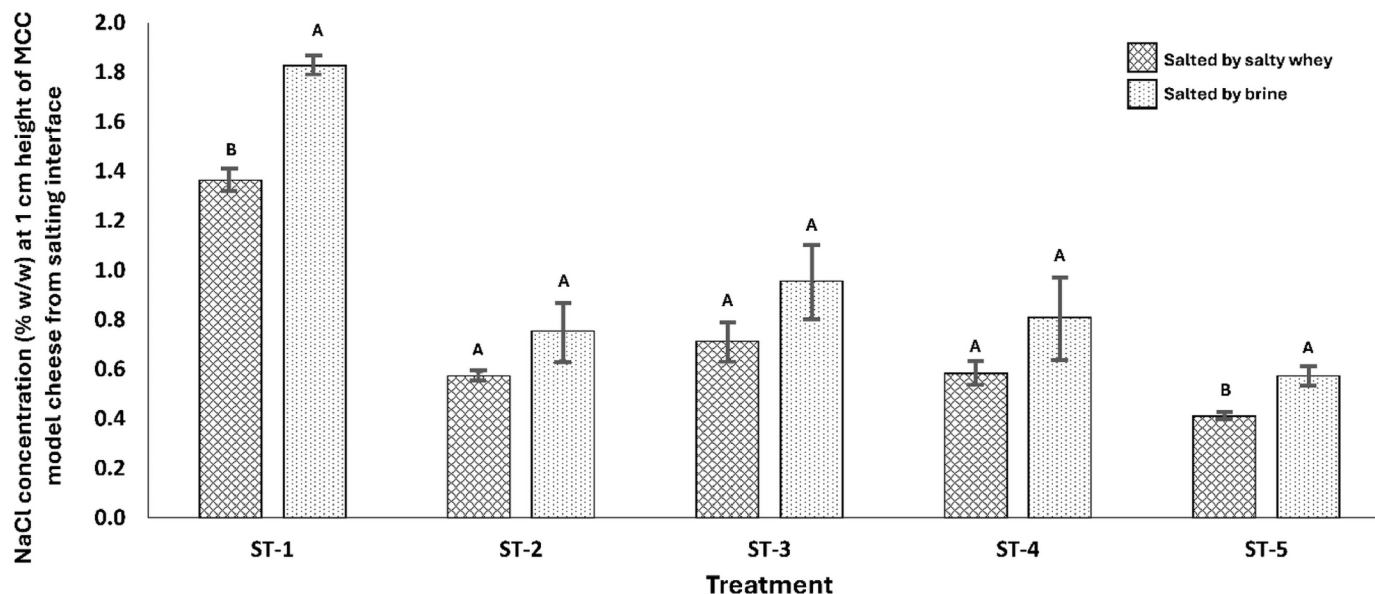


Figure 7. Comparison of salt diffusion in rennet-induced MCC gel samples (ST-1 to ST-5) when salted with brine versus salty whey: ST-1 = highest salt diffusion, ST-2 = lowest salt diffusion, ST-3 = moderate salt diffusion, ST-4 = control, ST-5 = high-fat sample. All samples were adjusted to similar pH, calcium, and NaCl concentrations. Values are expressed as mean $\pm$ SD for 3 replicates; same treatments having different uppercase letters (A,B) are significantly different ($P < 0.05$); ST-2, ST-3, and ST-4 were not significantly different.

to compositional differences in the media. Salty whey contains a complex mixture of lactose, lactate, whey proteins, and minerals, most notably NaCl (3.91% wt/wt) and KCl (1.08% wt/wt), which account for over 50% of its total mineral content, along with calcium phosphate and other minor components (Nishanthi et al., 2017).

The presence of KCl and lactose in salty whey increases the osmotic pressure and viscosity of the solution. Elevated osmotic pressure reduces water activity, thereby decreasing the driving force for salt migration. Concurrently, increased viscosity hinders ionic mobility, reducing the inward diffusion of Na^+ and Cl^- ions into the MCC matrix. The viscous nature of salty whey likely restricts mass transfer and disrupts the continuity of salt diffusion channels. Although treatments ST-2, ST-3, and ST-4 did not show significant differences in salt uptake between the 2 media, the trend remained consistent; brine outperformed salty whey in promoting salt migration. These results suggest that although salty whey may serve as a partial alternative to conventional brine, its effectiveness is reduced by physicochemical constraints.

CONCLUSIONS

Calcium concentration and salting temperature had a significant impact on the salt diffusion rate. Specifically, adding 1% (w/w) CaCl_2 or salting at 30 °C enhanced salt uptake in the MCC gel matrix. Textural and microstructural analyses confirmed that calcium facilitated salt

migration due to a more continuous and compact protein network. In contrast, increased fat content hindered salt diffusion due to obstruction of ion pathways. The FTIR analysis showed that samples with higher salt migration exhibited more stable and ordered protein secondary structures characterized by hydrogen bonds. Salty whey was less effective as a salting medium than standard brine, most probably due to elevated osmotic pressure and viscosity associated with KCl and lactose. Overall, the present study demonstrated that micellar casein-based cheese is a useful, controllable system for studying salt migration having a potential to serve as a versatile platform to model industrial cheese matrices.

NOTES

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[.org/10.6084/m9.figshare.30052393](https://doi.org/10.6084/m9.figshare.30052393). No human or animal subjects were involved and therefore approval from an Institutional Animal Care Committee or Institutional Review Board was not required. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: *D* = salt diffusion coefficient; FTIR = Fourier transform infrared; GDL = glucono- δ -lactone; ICP-OES = inductively coupled plasma optical emission spectroscopy; MCC = micellar casein concentrate; NR = not relevant; NS = not selected for further analysis; S/M = salt in moisture; S/N = serial number; ST = selected treatment; ST-1 = highest salt diffusion; ST-2 = lowest salt diffusion; ST-3 = moderate salt diffusion; ST-4 = control; ST-5 = high-fat sample.

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